用教师变化的一致性，筛选给学生的学习信号

Waypoint-OPD: Transferring RL-Induced Policy Shifts via Gated On-Policy Distillation

Bingxi Hou, Guochao Jiang, Guofeng Quan, Weiqing Li, Wenfeng Feng, Guohua Liu, Yuewei Zhang

论文标识：arXiv:2610.21601v1　建议阅读：45 分钟

核心观点

只看教师训练前后的变化，无法辨别中途是否发生反向变化。作者用前后两段变化的一致性给蒸馏信号加权；在所测数学与代码设置中取得更高领域平均分，但并非每个单项都胜出。

先修知识

Policy / RL：下一token的概率分布；奖励反馈改变生成偏好。
OPD → Direct-OPD：学生自己生成，由教师反馈或教师训练前后变化提供学习信号。

Policy / RL 入门
OPD 基础论文
Direct-OPD 直接前序

阅读路径与原文入口

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05 边界 / 5 分钟
方向一致，不等于老师一定正确
PDF第4页 · 正确性限制　PDF第8页 · 归因限制

实测：实验直接支持；推导：公式或数学关系；作者主张：作者判断；导读解释：辅助说明。
入口页码指原论文PDF页序；印刷页码以正文页脚为准。

WAYPOINT-OPD: TRANSFERRING RL-INDUCED POLICY SHIFTS VIA GATED ON-POLICY DISTILLATION

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ABSTRACT

Policy-shift distillation reuses the difference between a teacher's log probabilities before and after reinforcement learning (RL) as supervision on student-generated trajectories. However, this endpoint policy shift cannot reveal whether local action preferences evolved consistently during teacher training. The same endpoint shift can result from early and late shifts that reinforce or partially cancel each other. We investigate whether agreement between early and late teacher policy shifts can guide selective transfer. We introduce **Waypoint-guided On-Policy Distillation**(Waypoint-OPD), which uses an intermediate teacher checkpoint from the same RL run to compare early and late shifts in relative action preferences at student-visited prefixes. We gate the endpoint shift according to the directional agreement between the early and late shifts, suppressing it when they point in opposing directions. Compared with endpoint-only transfer, our method requires only one additional frozen-teacher scoring pass. Using RL-trained Qwen3-4B teachers, we evaluate our method with Qwen3-8B, Qwen3-14B, and Qwen3-30B-A3B as student models in mathematical and code reasoning. In all six domain–student settings, Waypoint-OPD achieves the highest domain-average score among the compared distillation methods. Ablations with the Qwen3-8B student show that removing or shuffling gates reduces mathematical reasoning performance. These results support the use of directional agreement between early and late teacher policy shifts to weight the endpoint shift during student training. Our code is available at https://github.com/hhhbxloveai/WaypointOPD.

student-generated trajectories：学生自己生成的整段回答序列。

【解读・译句】相同的起终点变化，可能来自前后两段相互增强、或部分抵消的变化。

student-visited prefixes：题目加上学生已生成的部分；老师据此为下一词打分。

【主张・要点】指两个领域各三种学生的领域平均。数学表有两处基准回落；这里没有逐题结果。

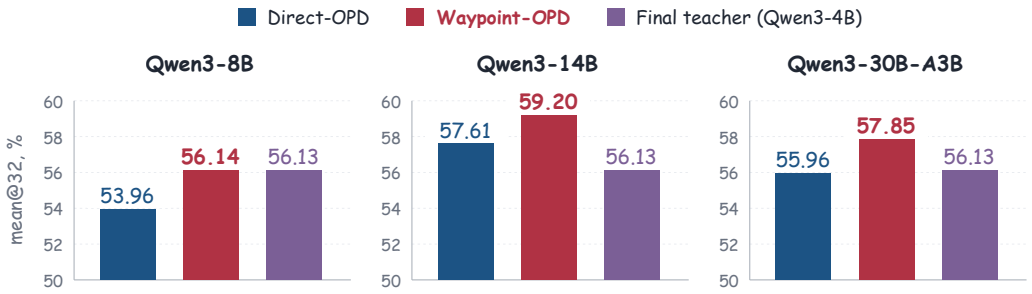


Figure 1: **Mathematical reasoning across three student models.** mean@32 accuracy (%), averaged over seven benchmarks. Waypoint-OPD improves over Direct-OPD by 2.18, 1.59, and 1.89 percentage points, respectively. Teacher scores are from the Final Qwen3-4B checkpoint.

【解读・图解】图1比较三种学生的数学平均，不是逐题胜负。

1 INTRODUCTION

Reinforcement learning can improve language-model instruction following using rewards learned from human preferences (Ouyang et al., 2022), and mathematical and code reasoning using verifiable

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rewards (Shao et al., 2024; Guo et al., 2025; Yu et al., 2025). The latter requires repeated generation, verification, and optimization of on-policy responses. Reusing a teacher's RL improvements offers a way to share this training effort across target models. This raises the question: which local changes in the teacher's policy should guide a student on its own generated trajectories?

Standard on-policy distillation (OPD) trains the student to match the teacher's token distributions on student-generated prefixes (Agarwal et al., 2024). Its supervision remains tied to the teacher's absolute policy and capability limitations. On-Policy Delta Distillation (OPD^2) uses the change between a reasoning-tuned teacher and its base model, retaining token-level signals when the centered change agrees in sign with the centered standard distillation signal (Heo et al., 2026). Direct On-Policy Distillation (Direct-OPD) focuses on RL-induced improvements. It uses the log-probability ratio between pre-RL and post-RL teacher checkpoints as a dense implicit reward on student-generated states, transferring the teacher's policy change without matching its absolute distribution (Feng et al., 2026). Both methods extract changes from a pair of teacher models, but neither directly measures whether consecutive changes during teacher RL point in compatible directions.

The Initial-to-Final log-probability ratio captures only the change between the teacher's policies before and after RL training. The same endpoint signal can arise from early and late policy changes that point in similar or opposing directions. Endpoint-only transfer cannot distinguish these cases. We hypothesize that agreement between the early and late changes provides evidence for deciding how much weight to assign an endpoint shift at a student-generated prefix.

【主张・要点】这是待检验的假设：教师前后两段变化方向一致，可作为终点变化该赋多大权重的线索。

Building on this hypothesis, we propose Waypoint-OPD, which uses agreement between consecutive teacher shifts to gate the endpoint transfer signal. We retain three checkpoints from the same teacher RL run: Initial before RL, Waypoint at an intermediate RL step, and Final at the end of RL. At each student-generated prefix, we select its top-k most probable next-token candidates. Over these candidates, we compare how the teacher's relative preferences change from Initial to Waypoint and from Waypoint to Final. Their agreement gates the centered Initial-to-Final signal, retaining aligned changes and suppressing reversals. Waypoint thus supplies evidence for weighting the endpoint shift, which remains the transferred signal. Relative to endpoint-only transfer, this requires one additional frozen teacher scoring pass, with no new teacher training or verifier calls during student transfer.

【解读・机制】两段变化的一致性控制去均值后的起终点信号：保留同向变化，抑制反向变化。
限制：一致性是加权线索，不保证监督正确。

Our contributions are:

- **Temporal agreement as a criterion for selective transfer.** We use the Waypoint checkpoint to assess temporal agreement at student-generated prefixes, revealing directional information unavailable from the endpoint pair alone.
- **Waypoint-guided gating of endpoint signals.** We develop Waypoint-OPD, a dense on-policy objective that uses this agreement to weight the endpoint signal at each student-generated prefix, with one additional frozen teacher scoring pass.
- **Higher average performance across domains and model sizes.** We evaluate Qwen3-8B, Qwen3-14B, and Qwen3-30B-A3B(Yang et al., 2025) on mathematics and code. Waypoint-OPD achieves the highest recorded domain average among the evaluated baselines in all six settings (Figure 1 and Tables 1 and 2). Ablations on Qwen3-8B mathematical reasoning support the value of temporal gating (Table 3).

2 PRELIMINARIES

2.1 PROBLEM SETUP AND NOTATION

Let $x \sim \mathcal{D}$ be a prompt, $y = (y_1, \ldots, y_T)$ a response, and $s_t = (x, y_{<t})$ the prefix at which a candidate token a is scored. The student π_θ generates on-policy responses $y \sim \pi_\theta(\cdot \mid x)$. Its initialization π_S remains fixed as a reference during transfer.

We denote the policies of the Initial, Waypoint, and Final teachers by π_0, π_m, and π_1, respectively. All three checkpoints come from the same teacher RL run: Initial before RL, Waypoint at an intermediate step, and Final after RL. They remain frozen during transfer. We measure the teacher's RL-induced change relative to π_0 and regularize student updates relative to student initialization π_S. Teachers and student share a tokenizer, allowing direct comparison of probabilities for the same token.

【主张・要点】这里需要同一词表下可对齐的 token；跨词表迁移不在此设置内。

https://arxiv.org/abs/2610.21601v1 | CC-BY-4.0 | notes added | source p.2

2.2 On-Policy Distillation

On-policy distillation (OPD) samples responses from the student and aligns its next-token distribution with the teacher's at the prefixes the student visits (Agarwal et al., 2024). Using the Final teacher π_1, OPD minimizes the reverse-KL objective

reverse-KL objective：以学生分布为第一项的KL目标，让学生接近老师。

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi_\theta(\cdot|x)}\left[\sum_{t=1}^{T} D_{\text{KL}}(\pi_\theta(\cdot \mid s_t) \,\|\, \pi_1(\cdot \mid s_t))\right]. \tag{1}$$

2.3 Direct On-Policy Distillation

Direct-OPD uses the teacher's RL-induced policy shift as an implicit reward on student-generated responses (Feng et al., 2026). The Initial-to-Final shift decomposes into token-level log-probability ratios:

$$\Delta^{(0,1)}(y \mid x) = \log \frac{\pi_1(y \mid x)}{\pi_0(y \mid x)} = \sum_{t=1}^{T} \delta_t(y_t), \tag{2}$$

【主张・要点】学生不是照着中间老师学答案；中间点只帮助判断终点变化该有多大权重。

where

$$\delta_t(a) = \log \pi_1(a \mid s_t) - \log \pi_0(a \mid s_t). \tag{3}$$

A positive shift indicates that teacher RL increased a token's probability at s_t; a negative shift indicates a decrease. At an idealized teacher optimum with KL regularization toward π_0, the sequence log-ratio recovers the reward up to positive scale and a prompt-dependent constant (Rafailov et al., 2023) as used by Feng et al. (2026). This motivates the endpoint signal; we do not assume that finite-step teacher RL attains this optimum.

【推导・公式】去均值：每个候选的变化减去按学生概率加权的平均变化，去掉不改变相对偏好的共同偏移。
符号：z(a)：候选词a上的变化；q(a)：学生在候选集上的归一化概率；E_q[z]：按q加权的平均变化；C_q[z](a)：z(a)减去该平均。
设计：统一加一个与候选词无关的常数不改变相对偏好，却可能改变原始变化向量的夹角；去掉这一分量后再比较方向。
例子：原页15附录A.1的恒等式：C_q[z+b]=C_q[z]，其中b与候选词无关；因此任意统一常数偏移都被去掉。
限制：仅在固定学生前缀、候选集和权重下讨论该关系，不能外推训练全过程不变。

The idealized sequence-level objective maximizes this shift while regularizing the student toward its own initialization:

$$\mathcal{J}_{\text{Direct-OPD}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\Big[\mathbb{E}_{y\sim\pi_\theta(\cdot|x)}[\Delta^{(0,1)}(y \mid x)] - \beta D_{\text{KL}}(\pi_\theta(\cdot \mid x) \,\|\, \pi_S(\cdot \mid x))\Big], \tag{4}$$

where $\beta > 0$ controls the student KL penalty. In practice, Direct-OPD uses $\delta_t(a)$ as an immediate token-level reward at student-generated prefixes, without verifier queries.

3 Method

Waypoint-OPD augments the endpoint signal in Section 2.3 with the Waypoint checkpoint π_m from the same teacher RL run. Agreement between the early and late shifts weights the endpoint signal on student-visited prefixes. Waypoint supplies consistency evidence rather than an imitation target. Figure 2 shows the pipeline.

3.1 Student-local trajectory consistency

At a student-visited prefix s_t, $\mathcal{K}_t = \text{TopK}(\pi_\theta(\cdot \mid s_t), k)$ selects the k token candidates with the highest student probabilities. The normalized student distribution on this support is

$$q_t(a) = \frac{\pi_\theta(a \mid s_t)}{\sum_{b\in\mathcal{K}_t} \pi_\theta(b \mid s_t)}, \qquad a \in \mathcal{K}_t. \tag{5}$$

These weights focus the comparison on likely student actions. Consistency is evaluated on the local support that receives the student update, with more probable candidates receiving greater weight. We score these candidates with all three frozen teachers. The two consecutive teacher shifts are

$$\delta_t^{(1)}(a) = \log \pi_m(a \mid s_t) - \log \pi_0(a \mid s_t), \tag{6}$$

$$\delta_t^{(2)}(a) = \log \pi_1(a \mid s_t) - \log \pi_m(a \mid s_t), \tag{7}$$

【解读・译句】各候选加上同一个与动作无关的常数时，相对偏好不变，但原始变化向量的夹角仍可能改变。

and telescope to $\delta_t = \delta_t^{(1)} + \delta_t^{(2)}$.

A uniform shift across candidates leaves their relative preferences unchanged, yet can affect the angle between raw shift vectors. We remove this action-independent component with the q_t-centering operator

$$C_{q_t}[z](a) = z(a) - \mathbb{E}_{b\sim q_t}[z(b)] \tag{8}$$

https://arxiv.org/abs/2610.21601v1 | CC-BY-4.0 | notes added | source p.3

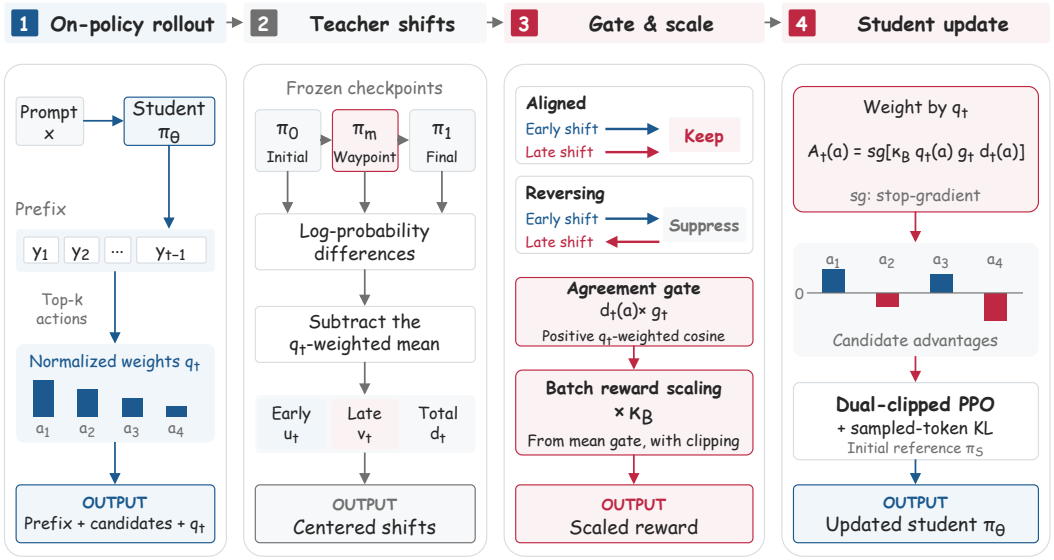


Figure 2: **Training pipeline of Waypoint-OPD.** Initial, Waypoint, and Final score candidates at student-generated prefixes. Their centered segment shifts define an agreement gate on the endpoint shift. Batch scaling and detached candidate advantages drive the student update Waypoint measures consistency only.

【主张・图解】图2：学生生成候选→三个教师打分→一致性门控与缩放→更新学生。

and write $u_t = C_{q_t}[\delta_t^{(1)}]$, $v_t = C_{q_t}[\delta_t^{(2)}]$, and $d_t = C_{q_t}[\delta_t] = u_t + v_t$. We weight candidates by student probability using the inner product $\langle u_t, v_t \rangle_{q_t} = \sum_{a \in \mathcal{K}_t} q_t(a) u_t(a) v_t(a)$. The local temporal agreement and weighted norms are

$$c_t = \frac{\langle u_t, v_t \rangle_{q_t}}{\|u_t\|_{q_t} \|v_t\|_{q_t}}, \quad \|u_t\|_{q_t} = \sqrt{\langle u_t, u_t \rangle_{q_t}}, \quad \|v_t\|_{q_t} = \sqrt{\langle v_t, v_t \rangle_{q_t}}. \tag{9}$$

【推导・公式】门控：u、v是老师前后两段去均值变化；c是按学生概率加权的夹角余弦；g=max(0,c)。同向保留，反向抑制。
符号：u、v：两段变化；q：学生在候选词上的归一化概率；c：加权余弦；g：门值；ε：低范数阈值。
设计：先去均值以去掉不改变候选相对偏好的统一偏移；按q加权关注学生可能选的词。余弦比较方向，max(0,c)抑制反向信号。任一范数太小时回退g=1。
例子：原页4自带例子：同一个非零总变化d，u=v=d/2时c=1、g=1；u=2d、v=-d时c=-1、g=0。两者终点相同，门值不同。
限制：方向一致不保证监督正确；低范数回退不是设为0。

3.2 Agreement Gating and Batch-Scale Normalization

When both centered segment norms exceed $\varepsilon = 10^{-8}$, we use the positive-cosine gate

$$g_t = \max(0, c_t). \tag{10}$$

If either norm is at most ε, we use the low-norm fallback $g_t = 1$. The gate $g_t \in [0, 1]$ is shared by all candidates at prefix s_t. Outside this fallback, it removes the endpoint signal when $c_t \leq 0$ and scales its magnitude by c_t otherwise. Algorithm and implementation conventions are given in Appendix B.

For illustration, consider two decompositions of a nonzero q_t-centered endpoint d, both outside the low-norm fallback. The pair $u = v = d/2$ gives $c = 1$ and $g = 1$, whereas $u = 2d$, $v = -d$ gives $c = -1$ and $g = 0$. Both yield $u + v = d$, so endpoint-only transfer cannot distinguish them. Appendix A.1 derives the endpoint's independence from Waypoint at fixed student prefixes and weights. Directional agreement alone does not guarantee beneficial supervision.

To stabilize the average gate scale across batches, we multiply all gated rewards in batch B by

$$\kappa_B = \mathrm{clip}\left(\frac{\tau}{\max\left(|B_{\text{pref}}|^{-1} \sum_{t \in B_{\text{pref}}} g_t, \varepsilon_{\text{num}} \right)}, \kappa_{\min}, \kappa_{\max} \right), \tag{11}$$

where B_{pref} contains valid response prefixes in B, and $\varepsilon_{\text{num}} > 0$ is a numerical floor preventing division by zero. The ratio targets a mean rescaled gate of $\tau > 0$, while clipping restricts κ_B to $[\kappa_{\min}, \kappa_{\max}]$, with $0 < \kappa_{\min} \leq \kappa_{\max}$. This shared positive multiplier preserves relative gate weights and zero gates. The effective multiplier $\kappa_B g_t$ can exceed one; batch scaling does not fix the reward-vector norm.

【推导・要点】某段变化太小、方向不可靠时，门值回退为 1，保留信号；不是设为 0。

【主张・要点】方向一致只是一条筛选依据，不代表老师建议一定正确。

【推导・公式】批量缩放κ：目标均值τ除以批内平均门值，再裁剪到上下限。统一正倍率保留门之间比例和零门。
符号：κ：全批共享倍率；τ：目标平均门值；B：有效回答前缀；εnum：分母数值下限；clip：上下限裁剪。
设计：门值均值随批次变化，统一缩放调整信号总体尺度；不改变相对门权重，也不固定奖励或梯度范数。
例子：原页17无门控例子：所有g=1，κ=clip(0.8,0.75,1.5)=0.8。
限制：数值下限与低范数阈值不同；裁剪生效时不保证均值恰好等于目标。

【解读・译句】这个共享的正数倍率保留门权重之间的比例，也让原本为零的门继续为零。

https://arxiv.org/abs/2610.21601v1 | CC-BY-4.0 | notes added | source p.4

3.3 DENSE CANDIDATE OBJECTIVE

The final coefficient for candidate action $a \in \mathcal{K}_t$ is

$$A_t(a) = \mathrm{sg}[\kappa_B \, q_t(a) \, g_t \, d_t(a)] \, . \tag{12}$$

Following the signal-minus-KL form of Eq. 4, the local on-policy surrogate to maximize is

$$\begin{aligned} \mathcal{J}_{\mathrm{WP}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\Bigg[\mathbb{E}_{y\sim\pi_\theta(\cdot|x)} \left[\sum_{t=1}^{T} \sum_{a\in\mathcal{K}_t} A_t(a) \log \pi_\theta(a \mid s_t) \right] \\ - \beta \, \widehat{D}_{\mathrm{KL}}(\pi_\theta(\cdot \mid x) \,\|\, \pi_S(\cdot \mid x)) \Bigg] , \end{aligned} \tag{13}$$

stop-gradient：At中门控、权重和终点信号不回传梯度；学生对数概率仍更新。

Rollout samples and candidate supports are held fixed when differentiating this surrogate, and the stop-gradient operator sg detaches all factors in A_t. Candidate credit is dense on $\mathcal{K}_t$, while the KL penalty is estimated on sampled response tokens and anchors the student to π_S. The implementation uses dual-clipped PPO, which has the same gradient as this surrogate at the rollout policy (Appendix B).

Relative to Direct-OPD, Waypoint-OPD adds one frozen teacher checkpoint and scoring pass, without additional trainable parameters or student rollouts. Derivations and signal properties appear in Appendix A.

4 EXPERIMENTS

We examine three questions:

Q1. Does Waypoint-OPD improve transfer across math, code, and model scales?

Q2. Does temporal consistency improve on endpoint transfer and Final-teacher imitation?

Q3. How do gate assignment, mean normalization, and Waypoint position affect performance?

Supplementary training-time and response-length analyses appear in Appendix H.

4.1 MODELS, DATA, AND EVALUATION

Teacher checkpoints and student models. For each domain, we use three checkpoints from the same Qwen3-4B (Yang et al., 2025) RL run: Initial (π_0) before RL, Waypoint (π_m) at RL step 600, and Final (π_1) at RL step 1200. Math and code teachers use non-thinking RL. All three checkpoints remain frozen during transfer. We train Qwen3-8B, Qwen3-14B, and Qwen3-30B-A3B students using the teacher's RL-induced policy shifts. These students span three scales and dense and MoE architectures within one family with compatible tokenizers. The transfer objective does not require the teacher to outperform the student. We report all three teacher checkpoints and the Base students in the result tables to show pre-transfer performance.

【解读・译句】该迁移目标并不要求教师的表现优于学生。

Training and evaluation data. Math transfer uses DAPO-Math-17K prompts (Yu et al., 2025). We evaluate on seven benchmarks: AIME 2024, AIME 2025, AIME 2026, HMMT February 2025, HMMT November 2025, HMMT February 2026, and AMC 2023. For each problem, `mean@32` measures average correctness over 32 generated responses. Benchmark scores average these values over problems, and Avg. is the unweighted mean across benchmarks.

【解读・说明】不是 32 次里至少成功一次；它先平均每题 32 次回答的正确性，再汇总。

Code transfer uses a Codeforces-derived prompt set. Evaluation uses the 378-problem MBPP+ set from EvalPlus (Liu et al., 2023), reporting both the original MBPP tests (Austin et al., 2021) and the original plus additional tests (MBPP+), together with LiveCodeBench v4, v5, and v6 (Jain et al., 2025). We report `mean@8`, which averages correctness over eight generated responses per problem and then over the problems in each benchmark. Code Avg. is the unweighted mean over these five evaluation scores. Within each benchmark, methods share prompt templates, decoding parameters, maximum generation lengths, and sample counts.

【解读・要点】两领域都用AIME24选检查点。整理者判断：数学表的AIME24列不是独立留出成绩；代码表不含该列，但沿用此选模依据。

Student checkpoint selection. For both domains, we use AIME 2024 as the validation set and select each trained student's checkpoint with the highest validation accuracy. All benchmark results for that student use this selected checkpoint.

https://arxiv.org/abs/2610.21601v1 | CC-BY-4.0 | notes added | source p.5

Implementation. We build on `verl` (Sheng et al., 2025) and score the student's top-16 token IDs with the three frozen teachers. The transfer objective combines the dense distillation reward with the student KL penalty. Answer graders are used only for evaluation. Implementation and training/evaluation configurations appear in Appendices B and C.

【主张・要点】老师的 RL 训练与学生迁移要分开；迁移时不额外调用判题器，不等于全流程从未使用。

4.2 Baselines and controls

We compare against the Base student, standard on-policy distillation (OPD) from the Final teacher (Agarwal et al., 2024), OPD^2 with its sampled-token objective (Heo et al., 2026), and endpoint Direct-OPD using $\log \pi_1 - \log \pi_0$ (Feng et al., 2026). All methods share training and evaluation settings, except for method-specific hyperparameters.

Three controls on Qwen3-8B math retain centering and alter one component of the full method. **No gate** sets $g_t = 1$ while retaining the gate-mean normalization rule. **Shuffled gate** permutes gates among valid response prefixes within a batch, retaining each prefix's candidate weights and centered endpoint signal. **No normalization** fixes $\kappa_B = 1$ while retaining the positive-cosine gate. Appendix E gives definitions and invariants of the shuffle control. We separately vary the math Waypoint over teacher RL steps 500, 600, and 700 on Qwen3-8B in Appendix F.

5 Results and Analysis

We compare transfer performance across math and code at three student scales (Section 5.1). We examine learning trajectories and performance relative to the Final teacher on Qwen3-8B math (Section 5.2). We assess the roles of temporal gating, gate assignment, and mean normalization through ablations (Section 5.3).

5.1 Main results across domains and scales

Tables 1 and 2 report math scores and code averages. Full code results appear in Appendix D. Waypoint-OPD achieves the highest domain-average score among the evaluated student methods in all six settings.

Math. Waypoint-OPD improves over Direct-OPD, the strongest average baseline for all three students, by 2.18, 1.59, and 1.89 percentage points for Qwen3-8B, Qwen3-14B, and Qwen3-30B-A3B, respectively. It improves 19 of 21 benchmark comparisons. The exceptions are HMMT25N for 14B (47.60 to 47.08) and HMMT26F for 30B-A3B (33.05 to 32.77). These results demonstrate improvements over endpoint transfer across student scales and both dense and MoE architectures. The 14B and 30B-A3B students also exceed the Final teacher's 56.13 average by 3.07 and 1.72 points, respectively, while 8B essentially matches it at 56.14. The 14B student surpasses the teacher on all seven math benchmarks.

【实测・发现】相对Direct-OPD，三种学生数学平均分别增加2.18、1.59、1.89个百分点。限制：21项比较有两项回落；不是每个基准均提升。

【实测・发现】14B学生在七项数学基准上均超过Final教师。限制：教师为Qwen3-4B；不代表任意学生、任务或教师配置。

Code. Waypoint-OPD improves over Direct-OPD by 2.66, 1.52, and 1.19 percentage points on average for Qwen3-8B, Qwen3-14B, and Qwen3-30B-A3B, respectively, with gains in all 15 code comparisons. OPD^2 achieves higher scores on 14B LCB v5 and on 30B-A3B MBPP and MBPP+, while Waypoint-OPD retains the highest average for each student (Appendix D). The 14B and 30B-A3B students also surpass the Final teacher's 55.37 average by 2.45 and 2.17 points, respectively, each scoring higher on four of the five code evaluations.

【实测・要点】数学比较有两处回落；“平均提升”不能改写成“全面胜出”。

【实测・发现】相对Direct-OPD，代码15项比较全部提升。限制：比较对象限定Direct-OPD；OPD2在三处单项更高。

Interpreting the endpoint comparison. The gains over Direct-OPD show that Waypoint-OPD improves transfer performance across both domains and all three student scales. OPD and OPD^2 provide complementary comparisons with Final-teacher imitation and selective distillation based on advantage sign agreement, respectively.

5.2 Learning dynamics

Figure 3 compares Waypoint-OPD and Direct-OPD at matched evaluation steps during Qwen3-8B math training. Unlike the selected-checkpoint results in Table 1, these trajectories show the performance gap over the same 100 updates.

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Table 1: **Mathematical reasoning across three student sizes.** `mean@32` scores (%) on seven benchmarks and their average. Bold marks the best score per student. Teacher checkpoints are references.

Method	AIME24	AIME25	AIME26	HMMT25 Feb	HMMT25 Nov	HMMT26 Feb	AMC23	Avg.
Teacher: Qwen3-4B								
Initial	21.67	21.67	21.67	12.50	10.00	15.15	71.25	24.84
Waypoint	58.96	54.79	56.77	31.35	35.73	31.72	93.44	51.82
Final	65.83	60.00	60.83	35.83	40.83	35.13	94.45	56.13
Student: Qwen3-8B								
Base	29.17	20.83	18.33	11.67	13.33	18.94	70.00	26.04
OPD	63.65	49.69	54.90	29.06	38.65	32.58	88.75	51.04
OPD^2	64.48	48.33	53.96	28.44	40.52	31.53	**92.97**	51.46
Direct-OPD	67.29	55.31	57.40	31.77	40.83	33.71	91.41	53.96
Waypoint-OPD	**68.85**	**57.92**	**60.21**	**34.38**	**42.60**	**36.36**	92.66	**56.14**
Student: Qwen3-14B								
Base	30.00	27.50	20.00	10.00	11.67	18.94	73.13	27.32
OPD	69.17	55.52	58.44	36.77	45.21	31.82	**96.25**	56.17
OPD^2	69.48	46.98	54.38	33.13	41.98	31.91	91.95	52.83
Direct-OPD	70.10	56.46	61.98	37.92	**47.60**	35.13	94.06	57.61
Waypoint-OPD	**70.21**	**60.52**	**63.75**	**40.21**	47.08	**36.55**	96.09	**59.20**
Student: Qwen3-30B-A3B								
Base	31.67	22.50	22.50	10.00	13.33	21.21	75.00	28.03
OPD	63.54	49.06	57.50	30.83	37.71	28.03	91.88	51.22
OPD^2	64.90	51.77	60.42	31.56	38.54	30.21	91.33	52.68
Direct-OPD	69.17	54.69	61.98	38.96	41.56	**33.05**	92.34	55.96
Waypoint-OPD	**70.94**	**58.44**	**63.65**	**39.90**	**44.69**	32.77	**94.53**	**57.85**

【实测・说明】Qwen3-8B学生数学平均：Direct-OPD 53.96 →Waypoint-OPD 56.14，增加2.18个百分点。

Table 2: **Average code reasoning performance.** Average `mean@8` scores (%) over five evaluations. Qwen3-4B teacher checkpoints are shared references. Bold marks the best score per student. Full results: Appendix D.

Teacher reference: Qwen3-4B **Initial: 37.64** **Waypoint: 52.22** **Final: 55.37**			
Method	Qwen3-8B	Qwen3-14B	Qwen3-30B-A3B
Base	41.03	47.72	50.33
OPD	43.91	51.66	51.91
OPD^2	51.21	55.60	55.96
Direct-OPD	50.59	56.30	56.35
Waypoint-OPD	**53.25**	**57.82**	**57.54**

【解读・说明】MBPP与MBPP+共享题目、测试不同。整理者判断：平均分中的二者不能当作独立题库。

Consistent gains across training checkpoints. Waypoint-OPD delivers consistent gains over Direct-OPD across both benchmarks and training checkpoints. Excluding the shared initialization, it scores higher in 13 of the 15 benchmark–checkpoint comparisons, with two exceptions early in training. From step 60 onward, Waypoint-OPD leads on all three benchmarks at every recorded evaluation, showing that its advantage is sustained through the end of training. The step-100 gains are 4.17, 3.13, and 0.10 percentage points on AIME 2025, AIME 2026, and HMMT November 2025, respectively. These results show that gains extend beyond the selected checkpoint to sustained advantages during training.

【实测・发现】训练轨迹15项比较中13项更高；从第60步起三个基准均领先。
限制：图3是8B数学三个基准，不代表所有模型与任务。

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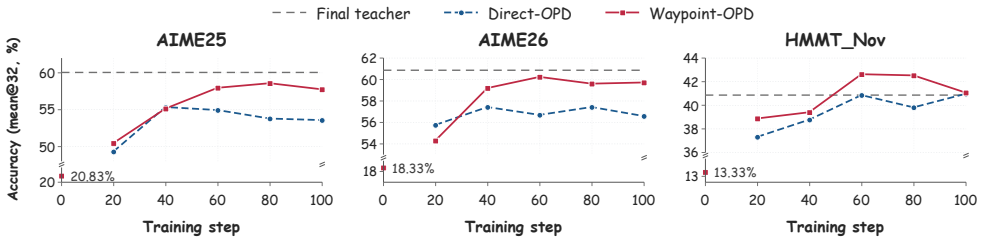


Figure 3: **Evaluation accuracy during training.** Qwen3-8B results on AIME 2025, AIME 2026, and HMMT November 2025 at steps 0, 20, 40, 60, 80, and 100. Scores are `mean@32` percentages. Solid red curves denote Waypoint-OPD and dashed blue curves denote Direct-OPD. Gray dashed lines show the Final Qwen3-4B teacher scores.

【解读・图解】图3看同一步数的曲线对比；灰线是教师参考，红线领先蓝线不等于超过灰线。

Comparison with the Final teacher. The gray reference lines distinguish gains over endpoint transfer from performance relative to the teacher. On HMMT November 2025, Waypoint-OPD exceeds the Final teacher's 40.83 at steps 60, 80, and 100, reaching 42.60 at step 60. On both AIME benchmarks, its recorded scores remain below the corresponding teacher reference. Improvements over Direct-OPD therefore coexist with different teacher–student orderings across benchmarks. The sustained HMMT advantage demonstrates that Waypoint-OPD can use the teacher's policy shifts to achieve performance beyond the Final teacher's score.

5.3 Gate and normalization ablations

Table 3: **Gate and normalization ablations on Qwen3-8B mathematical reasoning.** Bold marks the best score in each column.

Method	AIME24	AIME25	AIME26	HMMT25 Feb	HMMT25 Nov	HMMT26 Feb	AMC23	Avg.
No gate	66.25	54.27	56.67	32.60	39.79	34.66	90.23	53.50
Shuffled gate	63.33	52.92	54.58	30.00	37.08	31.06	89.45	51.20
No normalization	68.13	56.46	58.33	**34.79**	41.56	35.23	92.03	55.22
Direct-OPD	67.29	55.31	57.40	31.77	40.83	33.71	91.41	53.96
Waypoint-OPD	**68.85**	**57.92**	**60.21**	34.38	**42.60**	**36.36**	**92.66**	**56.14**

Removing the gate. On Qwen3-8B math, removing the gate lowers Avg. from 56.14 to 53.50, a decrease of 2.64 points, with lower scores on all seven benchmarks (Table 3). Retaining centering and normalization without consistency-dependent weighting therefore does not recover the full method's gain. This comparison does not isolate the effect of centering.

【主张・要点】去掉门控时仍保留去均值；这组实验不能单独证明去均值的贡献。

Disrupting the gate–prefix association. Shuffling gates lowers Avg. to 51.20, which is 4.94 points below the full method and 2.30 below the ungated variant, with the same ordering on every benchmark. Random reassignment is thus more detrimental than removing the selective weighting in this setting, supporting the value of matching each gate to the prefix whose temporal agreement produced it.

【实测・发现】打乱门控：平均51.20，比完整方法低4.94点，比无门控低2.30点。
限制：仅8B数学设置；不能把消融差值当作独立可加贡献。

Removing mean normalization. Fixing $\kappa_B = 1$ while retaining the positive-cosine gate lowers Avg. by 0.92 points to 55.22. Normalization improves six of seven benchmarks. The exception is HMMT February 2025 (34.79 without normalization versus 34.38 with it). The unnormalized variant still exceeds Direct-OPD by 1.26 points on average and on all seven benchmarks. Thus, the advantage persists without batch normalization, which provides an additional average benefit here. The remaining gain cannot be assigned entirely to gating because centering and other differences from Direct-OPD are also retained.

【解读・译句】剩余提升不能全归功于门控，因为去均值及相对 Direct-OPD 的其他差异也被保留。

https://arxiv.org/abs/2610.21601v1 | CC-BY-4.0 | notes added | source p.8

Ablation insights. These ablations support temporal weighting, its prefix assignment, and an incremental benefit from normalization in Qwen3-8B math. Their gaps are conditional on the retained components, not independent additive contributions. For a fixed rollout batch, shuffling preserves the gate distribution and κ_B, while advantage and gradient norms depend additionally on the gate–signal pairing (Appendix E). Supplementary diagnostics show stronger average gate attenuation in later batches, while mean-gate normalization does not fix the reward norm (Appendix G).

【主张・要点】各消融差值依赖保留下来的其他组件，不能把差值直接相加当作总贡献。

6 Related Work

On-policy distillation and policy-shift transfer. Knowledge distillation transfers teacher distributions or generated sequences (Hinton et al., 2015; Kim & Rush, 2016), while on-policy variants supervise student-generated trajectories (Agarwal et al., 2024). Complementary work refines divergence objectives and the use of teacher- and student-generated data (Wen et al., 2023; Gu et al., 2024; Ko et al., 2024; 2025). Weak-to-strong generalization asks whether weaker supervision can elicit stronger capabilities (Burns et al., 2024). WSPO and proxy-tuning reuse a weak model's adaptation contrast for training and decoding, respectively (Zhu et al., 2025; Liu et al., 2024). Within on-policy training, Direct-OPD transfers an RL-induced policy change through an endpoint log-ratio on student rollouts (Feng et al., 2026). W2S-OPD constructs a proxy teacher by adding a weak-model logit contrast to student base logits, then minimizes reverse KL on student rollouts. Its contrasts span RL checkpoints, model scales, and correct versus incorrect hints (Yu et al., 2026). OPRD uses a centered teacher–reference shift to amplify the component of a verifier-driven gradient along that direction (Park et al., 2026). Waypoint-OPD builds on Direct-OPD, weighting local endpoint signals by agreement between shifts before and after a Waypoint checkpoint.

Selective supervision in distillation. Teacher–student compatibility affects the utility of on-policy supervision (Li et al., 2026). Prune-OPD uses support overlap for suffix attenuation (Yang et al., 2026). EOPD uses teacher uncertainty in its entropy-dependent forward KL (Jin et al., 2026), while PW-OPSD uses token position in its reference-conditioned self-distillation (Liu et al., 2026). OPD^2 gates sampled-token updates by sign agreement between centered teacher–base and teacher–student advantages (Heo et al., 2026), whereas RG-OPD gates trajectories using agreement between verifier-derived advantages and teacher–student sequence log-likelihood gaps (Akhondzadeh et al., 2026). Waypoint-OPD uses different evidence: consecutive shifts in the teacher's own RL trajectory over student-supported candidate actions. Their weighted directional agreement produces a prefix-level gate for the dense endpoint update, without verifier feedback during transfer. This agreement is a weighting criterion, not a guarantee of correctness.

Checkpoint selection and progressive distillation. A teacher's final performance need not determine its teaching utility: early stopping and intermediate checkpoint selection can improve distillation (Cho & Hariharan, 2019; Wang et al., 2022). Pro-KD and progressive distillation use successive teacher checkpoints as targets or a curriculum (Rezagholizadeh et al., 2022; Panigrahi et al., 2025). Direct-OPD also studies RL checkpoint choice for transfer (Feng et al., 2026). For fixed Initial and Final endpoints, Waypoint measures directional consistency between the two shifts. It supplies neither an imitation target nor a curriculum. The centered Initial-to-Final shift remains the transferred object.

【解读・译句】它既不提供模仿目标，也不提供训练课程。

7 Conclusion

We presented Waypoint-OPD, which uses agreement between consecutive teacher shifts to transfer RL-induced policy changes. An intermediate Waypoint checkpoint reveals whether early and late changes in relative action preferences reinforce or oppose each other at student-generated prefixes. This information, unavailable from endpoint differences alone, determines the weight of the centered Initial-to-Final shift in the student's on-policy objective. The method requires one additional frozen-teacher scoring pass and no verifier calls during transfer. Experiments on mathematical and code reasoning show gains over endpoint transfer across three Qwen3 student scales. Ablations on Qwen3-8B math support prefix-specific gate assignment and batch-scale normalization. These findings suggest that intermediate teacher checkpoints provide evidence for weighting local policy changes during student learning.

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AI USE STATEMENT

We use generative AI tools for language polishing, assistance in clarifying mathematical derivations, and consistency checks of notation and exposition. These tools were not used for experimental design, implementation, or execution. We take full responsibility for the final content of this work.

ETHICS STATEMENT

Use of Human Annotations Human annotations are only used in methodological research at the beginning of the work, to assist in analyzing the feasibility of the proposed solution. Annotators consented to the use of data for research purposes. We ensure that the privacy of all annotators is protected throughout the annotation process, and all of them are adequately paid according to local standards. Human annotations are not applied during the evaluation of our method.

Risks In this paper, all datasets are obtained from official sources. The datasets adopted have been anonymized and do not contain offensive information. However, we cannot guarantee that the datasets do not contain socially harmful or toxic language.

REPRODUCIBILITY STATEMENT

The transfer objective is specified in Section 3, with supporting derivations and signal properties in Appendix A and pseudocode and optimization details in Appendix B. Section 4.1 describes the teacher checkpoints, student models, training and evaluation data, evaluation metrics, and student checkpoint selection. Appendix C reports hyperparameters, hardware, and evaluation settings.

Baselines and controls are described in Section 4.2, with precise definitions of the gate and normalization ablations in Appendix E and the Waypoint-position ablation setup in Appendix F. Per-benchmark results and domain averages appear in Tables 1 and 7, and gate and normalization ablation results appear in Table 3.

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A Derivations and Signal Properties

A.1 Endpoint preservation and gate properties

At fixed s_t, $\mathcal{K}_t$, and q_t, with finite teacher log scores, linearity of centering gives

$$u_t + v_t = C_{q_t}[\delta_t^{(1)} + \delta_t^{(2)}] = C_{q_t}[\log \pi_1 - \log \pi_0] = d_t. \tag{14}$$

Hence Waypoint affects g_t and κ_B, while the transferred endpoint d_t remains fixed. This comparison holds at fixed student prefixes and weights; subsequent training trajectories can differ.

【推导·要点】固定输入前缀与权重时，起终点信号不变；训练继续后，学生路径可以不同。

The remaining properties follow directly from the definitions. First, $C_{q_t}[z + b] = C_{q_t}[z]$ for any action-independent constant b, so such offsets in teacher scores leave the transfer signal unchanged. Second, outside the low-norm fallback, the gate is one for positively collinear shifts and zero for nonpositive weighted cosine. Independent positive rescaling of u_t and v_t leaves this gate unchanged provided neither segment enters the fallback; their sum d_t can change. Finally, $g_t \in [0, 1]$ and $\kappa_B > 0$ ensure that $A_t(a)$ cannot reverse the sign of $d_t(a)$. The effective multiplier $\kappa_B g_t$ can exceed one: batch normalization targets the mean gate, not a fixed reward norm. These are properties of the construction; the usefulness of temporal weighting is evaluated empirically.

A.2 Zero-sum coefficients and the local gradient

Centering under q_t immediately gives

$$\sum_{a \in \mathcal{K}_t} A_t(a) = \kappa_B g_t \sum_{a \in \mathcal{K}_t} q_t(a) d_t(a) = 0. \tag{15}$$

With $\mathcal{K}_t$ fixed and A_t detached, this cancels the common softmax term. Writing z_b for the student logit of vocabulary token b at s_t,

$$\frac{\partial}{\partial z_b}\left[-\sum_{a \in \mathcal{K}_t} A_t(a) \log \pi_\theta(a \mid s_t)\right] = \begin{cases} -A_t(b), & b \in \mathcal{K}_t, \\ 0, & b \notin \mathcal{K}_t. \end{cases} \tag{16}$$

The PPO surrogate has this gradient at the rollout policy (Appendix B.1). This local result excludes the KL term and does not imply that a shared-parameter update preserves candidate probability mass or leaves noncandidate probabilities unchanged.

【推导·要点】此局部结论不含KL项；不能推出共享参数更新会保住候选词总概率，或让非候选词概率不变。

B Algorithm and Implementation Details

Algorithm 1 Agreement-gated on-policy distillation of RL-induced policy shifts

Require: Prompts $\mathcal{D}$, student and reference policies π_θ, π_S, frozen Initial, Waypoint, and Final teachers π_0, π_m, π_1, hyperparameters k, τ, ε, scale bounds, and KL coefficient β.

for each update **do**
  Sample prompts x and responses $y \sim \pi_\theta(\cdot \mid x)$.
  for each valid prefix $s_t = (x, y_{<t})$ **do**
    Obtain $\mathcal{K}_t = \mathrm{TopK}(\pi_\theta(\cdot \mid s_t), k)$ and detached normalized weights q_t.
    Score $\mathcal{K}_t$ with π_0, π_m, and π_1.
    Form $u_t = C_{q_t}[\log \pi_m - \log \pi_0]$ and $v_t = C_{q_t}[\log \pi_1 - \log \pi_m]$.
    Compute c_t with Eq. 9 and set $g_t = \max(0, c_t)$. Use $g_t = 1$ if either centered norm is at most ε.
    Set $d_t = u_t + v_t$.
  end for
  Compute κ_B with Eq. 11.
  Set $A_t(a) = \mathrm{sg}[\kappa_B \, q_t(a) \, g_t \, d_t(a)]$.
  Update π_θ using the PPO surrogate in Eq. 17 and the KL penalty in Eq. 13.
end for

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B.1 CANDIDATE-WISE DUAL-CLIPPED PPO

We use the candidate-wise dual-clipped PPO surrogate (Schulman et al., 2017; Ye et al., 2020). Let π_{old} denote the student policy at rollout time, with $r_t(a) = \pi_\theta(a \mid s_t)/\pi_{\text{old}}(a \mid s_t)$, $\bar{r}_t(a) = \text{clip}(r_t(a), 1-\epsilon, 1+\epsilon)$, and let $c > 1$ be the dual-clip constant, distinct from c_t. For each candidate, the maximized surrogate is

$$J_t(a) = \begin{cases} \min\{r_t(a)A_t(a), \bar{r}_t(a)A_t(a)\}, & A_t(a) \geq 0, \\ \max\{\min[r_t(a)A_t(a), \bar{r}_t(a)A_t(a)], cA_t(a)\}, & A_t(a) < 0. \end{cases} \tag{17}$$

The loss sums $-J_t(a)$ over candidates and uses token-mean aggregation over valid response tokens, together with the sampled-token KL penalty to π_S in Eq. 13. At the rollout policy, $r_t(a) = 1$ and clipping is inactive, so maximizing $\sum_{a \in \mathcal{K}_t} J_t(a)$ gives the same local distillation update as Eq. 13 (Appendix A.2).

B.2 IMPLEMENTATION CONVENTIONS

Candidate supports $\mathcal{K}_t$ and detached coefficients $A_t(a)$ remain fixed during each student update. Batch-scale normalization uses valid response prefixes across the global rollout batch B. The low-norm fallback retains the endpoint signal when a segment has too little weighted variation to define a reliable direction, using $\varepsilon = 10^{-8}$. The normalization floor ε_{num} is the smallest positive normal float32 value ($\approx 1.18 \times 10^{-38}$), distinct from the low-norm threshold. Training and evaluation configurations appear in Appendix C.

【推导・要点】归一化分母的数值下限与低范数阈值是两回事，不能都设为同一个 epsilon。

C FULL EXPERIMENTAL PROTOCOL

We use `verl` (Sheng et al., 2025) with FSDP (Zhao et al., 2023) for training and vLLM (Kwon et al., 2023) for rollouts. Training and inference are performed on a single node with 8 NVIDIA H20 GPUs. Tables 4, 5, and 6 summarize the teacher and transfer training hyperparameters, model configurations, and evaluation benchmarks.

Teacher training. For math RL training, we select approximately 57K examples with difficulty levels ≥ 6 from DeepMath (DeepMath-103K) (He et al., 2026). For code RL training, we use Eurus-RL-Code (Cui et al., 2026), which contains approximately 25K training examples. We train both Qwen3-4B teachers with group relative policy optimization (GRPO) (Shao et al., 2024) for 1,200 optimization steps using the same hyperparameters in Table 4; only the training datasets differ.

Table 4: GRPO training configuration shared by the math and code teachers.

Setting	Value
Train batch size	128
Micro batch size	128
Rollout n	8
Maximum prompt length	2,048
Maximum response length	16,384
Temperature	1.0
Top-p	1.0
Learning rate	10^{-6}
Optimization steps	1,200
KL coefficient	0.0

Evaluation settings. Math and code evaluation use the same rollout settings: temperature 0.7, top-p 0.95, and a maximum generation length of 16,384 tokens. We report `mean@32` for math and `mean@8` for code, averaging correctness over 32 and eight responses per problem, respectively, and then over the problems in each benchmark. MBPP (Austin et al., 2021) and MBPP+ from EvalPlus (Liu et al., 2023) use the same problems with the original tests and the original plus additional tests, respectively. We also evaluate on LiveCodeBench v4, LiveCodeBench v5, and LiveCodeBench v6

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Table 5: Transfer configuration shared by math and code, except for the transfer datasets.

Setting	Value
Math training dataset	DAPO-Math-17K
Code training dataset	Codeforces-derived
Train batch size	256
Mini batch size	64
Rollout n	1
Maximum prompt length	2,048
Maximum response length	16,384
Temperature	1.0
Learning rate	10^{-6}
Optimization steps	100
KL coefficient β	2.5
Student support k	16
Low-norm threshold ε	10^{-8}
Target gate mean τ	0.8
Scale clip	$[0.75, 1.5]$

(Jain et al., 2025). For both domains, we select each trained student's checkpoint by its validation accuracy on AIME 2024 and report all benchmark results from that checkpoint.

Table 6: Model and evaluation configuration.

Item	Value
Initial teacher	Qwen3-4B
Waypoint teacher	RL step 600
Final teacher	RL step 1200
Students	Qwen3-8B, Qwen3-14B, Qwen3-30B-A3B
Math evaluation	AIME 2024, AIME 2025, AIME 2026, HMMT February 2025,HMMT November 2025, HMMT February 2026, AMC 2023
Code evaluation	MBPP, MBPP+, LiveCodeBench v4, LiveCodeBench v5,LiveCodeBench v6

D Full Code Results

Table 7 presents the complete code evaluation results corresponding to the averages summarized in Table 2 in the main text. It includes the three teacher checkpoints and all evaluated student methods.

E Component Ablation Details

We detail the three controls evaluated on Qwen3-8B mathematical reasoning in Table 3. All controls retain q_t-centering and the student KL penalty and use the training and evaluation settings in Appendix C. Each control collects rollouts from its current student policy.

No gate. This control sets $g_t = 1$ at every valid response prefix while retaining the centered endpoint signal and the gate-mean normalization rule in Eq. 11. With the default settings, the rule reduces to the constant multiplier

$$\kappa_B = \mathrm{clip}(0.8, 0.75, 1.5) = 0.8.$$

This control removes consistency-dependent weighting while retaining centering and the normalization rule.

【解读·说明】无门控对照仍按规则缩放；默认倍率是0.8，不能把它当成完全等同Direct-OPD。

Shuffled gate. We compute gates using the standard rule, including the low-norm fallback, and randomly permute them among valid response prefixes in the global rollout batch B. For a permutation

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Table 7: **Full code reasoning results across three student sizes.** Scores are `mean@8` percentages. Avg. is the unweighted mean across the five evaluations. Bold marks the best score within each student block. Teacher checkpoints are reference rows.

Method	MBPP	MBPP+	LCB v4	LCB v5	LCB v6	Avg.
Teacher: Qwen3-4B						
Initial	62.53	54.53	25.37	20.36	25.43	37.64
Waypoint	83.33	70.37	36.01	37.20	34.21	52.22
Final	88.00	72.65	38.61	40.57	37.00	55.37
Student: Qwen3-8B						
Base	67.16	58.63	30.82	23.35	25.21	41.03
OPD	69.58	57.67	35.64	26.95	29.71	43.91
OPD^2	79.10	68.68	36.54	37.72	34.00	51.21
Direct-OPD	78.14	65.81	36.63	37.95	34.43	50.59
Waypoint-OPD	**82.97**	**69.38**	**37.87**	**40.19**	**35.86**	**53.25**
Student: Qwen3-14B						
Base	83.86	71.69	30.94	26.95	25.14	47.72
OPD	84.49	72.62	35.64	33.53	32.00	51.66
OPD^2	85.98	72.92	38.61	**43.56**	36.93	55.60
Direct-OPD	85.32	73.58	43.44	42.96	36.21	56.30
Waypoint-OPD	**87.57**	**74.60**	**45.54**	43.11	**38.29**	**57.82**
Student: Qwen3-30B-A3B						
Base	86.34	73.18	32.55	28.67	30.93	50.33
OPD	86.61	72.25	35.15	32.49	33.07	51.91
OPD^2	**88.23**	**74.44**	40.10	41.62	35.43	55.96
Direct-OPD	87.24	73.77	42.33	41.54	36.86	56.35
Waypoint-OPD	88.00	74.21	**43.94**	**42.89**	**38.64**	**57.54**

σ, the advantage is

$$A_t^{\text{shuffle}}(a) = \text{sg}[\kappa_B q_t(a) g_{\sigma(t)} d_t(a)].$$

Candidate weights q_t and endpoint signals d_t remain at their original prefixes. For a fixed rollout batch, the permutation preserves the gate distribution and its mean, leaving κ_B unchanged. It breaks the association between each prefix and its own temporal-agreement weight. Advantage and gradient norms depend additionally on this gate–signal pairing.

【推导・要点】打乱保留门值分布和均值，但改变门与信号的对应；不保证优势或梯度范数不变。

No normalization. This control fixes $\kappa_B = 1$ while retaining centering, the positive-cosine gate, and the low-norm fallback:

$$A_t^{\text{unnorm}}(a) = \text{sg}[q_t(a) g_t d_t(a)].$$

For a fixed batch, removing the common multiplier preserves relative gate weights across prefixes while removing the batch-dependent rescaling of the distillation signal relative to the retained KL penalty.

F Sensitivity to Waypoint Checkpoint Selection

Experimental setup. We examine whether transfer performance depends on the precise position of the intermediate teacher checkpoint. On Qwen3-8B mathematical reasoning, we replace the default Waypoint at teacher RL step 600 with checkpoints at steps 500 and 700 from the same Qwen3-4B RL run. The Initial teacher before RL and the Final teacher at step 1200 remain fixed. The other training and evaluation settings follow the default Qwen3-8B math setup in Appendix C. Table 8 reports all seven benchmark scores and their unweighted average.

Small changes in aggregate performance. The average scores are 56.18, 56.14, and 56.13 for steps 500, 600, and 700, respectively, giving a total range of only 0.05 percentage points. Relative

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Table 8: **Sensitivity to the Waypoint checkpoint on Qwen3-8B mathematical reasoning.** Only the intermediate teacher checkpoint is varied, while the Initial and Final teachers are fixed. Step numbers refer to the teacher's RL training. Bold marks the best score in each column.

Waypoint	AIME24	AIME25	AIME26	HMMT25 Feb	HMMT25 Nov	HMMT26 Feb	AMC23	Avg.
Step 500	69.06	**58.13**	59.79	33.96	42.60	**36.74**	**92.97**	**56.18**
Step 600	68.85	57.92	**60.21**	34.38	42.60	36.36	92.66	56.14
Step 700	**69.27**	57.60	59.79	**34.58**	**42.81**	36.36	92.50	56.13

to the default step 600, moving the Waypoint earlier changes Avg. by $+0.04$ points, and moving it later changes Avg. by -0.01 points. All three choices exceed the Direct-OPD average of 53.96 by 2.17–2.22 points and outperform it on every individual benchmark (Table 1). Thus, the observed transfer advantage is preserved across these neighboring Waypoints.

Stability across Waypoint choices. Across the three tested Waypoints, the score range on each benchmark is at most 0.62 percentage points, with the largest range on HMMT25 February (33.96–34.58). Together with the 0.05-point range in Avg. and the consistent gains over Direct-OPD, these results show stable transfer performance across the tested neighboring Waypoints on Qwen3-8B math.

【实测·要点】中间点取500/600/700步时，数学平均56.18/56.14/56.13，相差0.05个百分点。
限制：只测8B数学的三个邻近中间点，不能推广为任意中间点均稳定。

G GATE DIAGNOSTICS

Figure 4 examines agreement gating and batch-scale normalization over 100 training steps of Waypoint-OPD on Qwen3-8B mathematical reasoning, using the configuration in Appendix C.

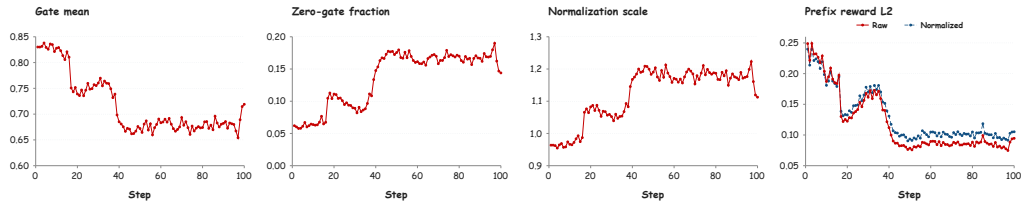


Figure 4: **Gate diagnostics on Qwen3-8B mathematical reasoning.** Statistics for Waypoint-OPD over 100 training steps. From left to right: mean gate g_t, zero-gate fraction, batch-scale normalization multiplier κ_B, and mean per-prefix reward-vector L2 norm. Raw uses the gated, student-weighted endpoint signal $q_t(a)g_t d_t(a)$; Normalized uses the final candidate coefficients $A_t(a)$ in Eq. 12 from the same run. All statistics aggregate valid response prefixes.

Temporal weighting over training. The mean gate g_t decreases from 0.810 in the first 20 steps to 0.683 in the last 20, while the zero-gate fraction increases from 7.26% to 16.59%. These summaries are unweighted averages of step-level statistics. The gate therefore applies stronger average attenuation before batch-scale normalization in later batches and removes the centered endpoint signal at a larger fraction of valid response prefixes. With the Initial, Waypoint, and Final teachers frozen, these statistics track temporal agreement on the evolving student-generated prefixes and candidate distributions.

【实测·发现】图4：平均门值从前20步0.810降至后20步0.683。
限制：反映演变中的学生前缀；教师本身冻结，不是教师继续训练。

Batch-scale normalization. The multiplier κ_B ranges from 0.955 to 1.223, remaining within the configured bounds $[\kappa_{\min}, \kappa_{\max}] = [0.75, 1.5]$. It is below one for the first 16 steps and above one thereafter, maintaining the mean rescaled gate at the target $\tau = 0.8$ to numerical precision. As specified in Eq. 11, this shared positive multiplier preserves relative gate weights and zero gates. It also scales each prefix's reward-vector L2 norm by κ_B. Over the last 20 steps, the mean per-prefix norms average 0.0855 before scaling and 0.1002 after scaling. Both norms decrease overall during training, while the mean rescaled gate remains stable. These diagnostics show how batch-scale

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normalization maintains the average gate scale as temporal weighting adapts to student-generated prefixes. The performance benefit is reflected in Table 3: mean normalization improves the Qwen3-8B math average from 55.22 to 56.14.

H Training Time and Response Length

Table 9 compares the total training time over 100 training steps of Qwen3-8B on math using eight H20 GPUs, excluding evaluation. Both methods follow the configuration in Appendix C. Waypoint-OPD completes training in 8 h 2 min 21 s, compared with 8 h 21 min 3 s for Direct-OPD, reducing total training time by 18 min 42 s (3.73%). The corresponding mean step times are 289.41 s and 300.63 s, a reduction of 11.22 s per step. Total GPU-hours decrease from 66.81 to 64.31.

Table 9: **Training-time comparison on Qwen3-8B math.** Total training time over 100 training steps on eight H20 GPUs, excluding evaluation.

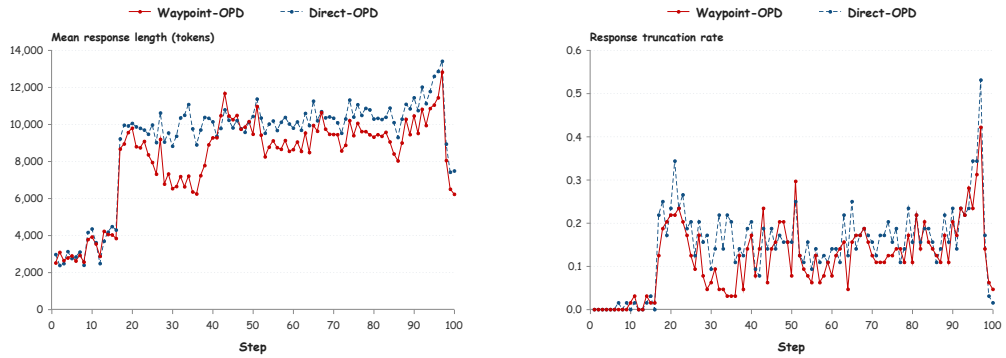
Method	Time (h:m:s)	Seconds/step	GPU-hours
Direct-OPD	8:21:03	300.63	66.81
Waypoint-OPD	8:02:21	289.41	64.31

GPU-hours are computed as eight times the reported training duration in hours and exclude evaluation.

Relative to Direct-OPD, Waypoint-OPD adds one scoring pass with the frozen Waypoint checkpoint while using the same number of student rollouts. With this additional source of temporal consistency information, Waypoint-OPD achieves a lower measured training time in this setting.

Figure 5 compares rollout generation over the same 100 training steps. Response length is the mean number of tokens per rollout response at each training step, and response truncation rate is the fraction of rollout responses reaching the maximum length of 16,384 tokens. Waypoint-OPD has a lower mean response length at 87 of the 100 steps and a lower response truncation rate at 62 steps, with equal rates at 18 steps. Averaged uniformly across the 100 steps, the response truncation rate decreases from 14.83% for Direct-OPD to 11.97% for Waypoint-OPD, a reduction of 2.86 percentage points. These results show that Waypoint-OPD produces shorter responses at most training steps and reaches the response-length limit less frequently on average.

【实测・发现】图5：100步中87步回答更短。
限制：未隔离回答长度对总耗时的因果贡献。



(a) Mean response length

(b) Response truncation rate

Figure 5: **Response length and response truncation rate.** The two panels compare Qwen3-8B trained with Waypoint-OPD (solid red) and Direct-OPD (dashed blue) over 100 training steps. (a) Mean number of tokens per rollout response at each training step. (b) Fraction of rollout responses reaching the maximum length of 16,384 tokens; 0.2 corresponds to 20%.

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